

Name _____ Surname _____ Date ____/____/____

Mock test — Speed, energy and work

Formative test, not graded. Time: 60 minutes. Calculator and formula sheet (last page) allowed. Use $g = 10 \text{ N/kg}$. Write on separate sheets, show every step of your working and write the unit in each answer. Give answers to 3 significant figures where needed. Total: 45 marks.

🏊 **Lifebuoy.** The parts marked with the lifebuoy are worth 27 marks out of 45, that is 60%: if you have completed all of them, you are safe. The other parts take you beyond the pass mark.

Part A — Quiz and concepts

Exercise 1 — Multiple choice

🏊 Only one answer is correct. Each question is worth 1 mark, 8 marks in total.

1. The energy stored in a stretched rubber band is:
A chemical B elastic (strain) C kinetic D nuclear
2. A car travels 150 km in 2.0 h. Its average speed is:
A 0.013 km/h B 300 km/h C 75 km/h D 152 km/h
3. On a distance–time graph, a horizontal line shows an object that is:
A speeding up B slowing down C moving backwards D stationary
4. A ball falls from a shelf. Ignoring air resistance, which quantity stays constant during the fall?
A total energy B kinetic energy C gravitational potential energy D speed
5. The speed of a trolley doubles. Its kinetic energy:
A doubles B stays the same C halves D becomes four times bigger
6. In which situation is work being done?
A a girl lifts a box onto a shelf B a boy holds a heavy bag still C a book rests on a table
D a child sits on a chair
7. One joule is equal to:
A 1 N/m B 1 Nm C 1 N/s D 1 kg m
8. A skier at the top of a slope has 80 kJ of gravitational potential energy. At the bottom, 30 kJ has been transferred to thermal energy. Her kinetic energy at the bottom is:
A 110 kJ B 80 kJ C 50 kJ D 30 kJ

Exercise 2 — True or false

- 🏊 Write T or F. No explanation needed. Each statement is worth 1 mark, 4 marks in total.
- (a) A steeper line on a distance–time graph means a lower speed.
 - (b) Energy is created in a power station and destroyed in a lamp.
 - (c) A book is pushed along a horizontal shelf: its gravitational potential energy stays the same.
 - (d) When a ball is thrown upwards, its kinetic energy decreases and its gravitational potential energy increases.

Exercise 3 — Energy stores

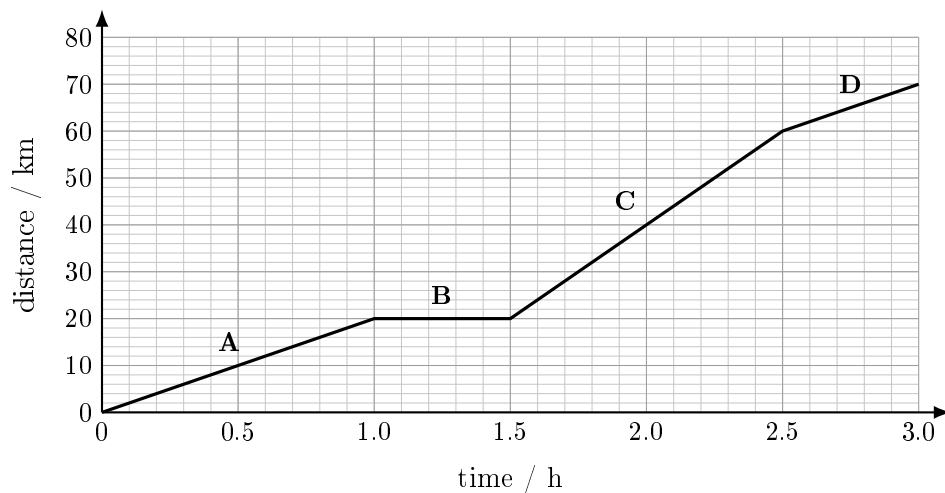
⚽ Complete the table. In each event, name the energy store that decreases and the one that increases. 1 mark for each row, 3 marks in total.

Event	Store that decreases	Store that increases
A boy lets go of a stretched catapult and the stone shoots out.		
A diver climbs the ladder to the 10 m board.		
A ball falls from a shelf to the floor, just before it hits.		

Part B — Problems

Exercise 4 — A cycling journey

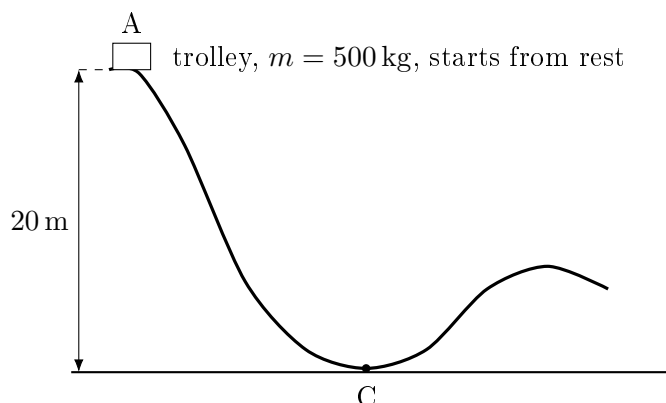
The graph shows the journey of a cyclist. It has four sections, A, B, C and D.



- (a) ⚽ Which section shows the cyclist stopped? Which section shows the highest speed? Write the letters. [2]
- (b) Calculate the speed in section C, in km/h. [2]
- (c) Calculate the average speed for the whole journey, in km/h. [2]
- (d) Write the speed of section C in m/s. [2]

Exercise 5 — The roller coaster

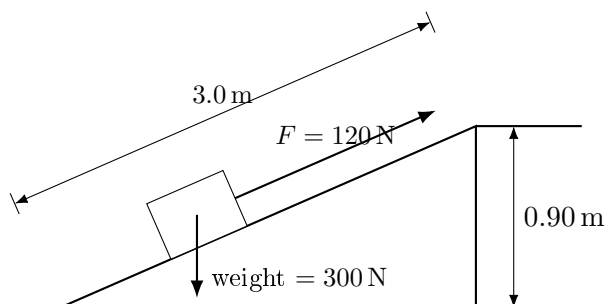
A trolley of mass 500 kg starts from rest at A and runs down the track to C. Ignore friction and air resistance.



- (a) Describe the energy transfer from A to C, naming the energy stores. Which principle tells you how the two amounts compare? [2]
- (b) Write, in symbols and without numbers, the equation that links the speed v at C to the height h of A. [2]
- (c) Calculate the gravitational potential energy lost by the trolley between A and C. [2]
- (d) Calculate the speed of the trolley at C. [2]

Exercise 6 — Pushing a box up a ramp

A girl pushes a box up a ramp with a force of 120 N parallel to the ramp. The box has a weight of 300 N.



- (a) Two lengths are marked on the diagram. Which one goes into $W = Fd$ and which one goes into $\Delta E_p = mgh$? Write the answer next to each formula. [2]
- (b) Calculate the work done by the girl on the box. [2]
- (c) Calculate the gain in gravitational potential energy of the box. [2]
- (d) The work done is greater than the energy gained by the box. In which energy store does the difference end up, and which force is responsible? [2]

Exercise 7 — Stopping a car

A car of mass 1200 kg travels at 15 m/s. The driver brakes and the brakes exert a constant force of 6000 N until the car stops.

- (a)  Write, in symbols, the equation that links the work done by the brakes to the kinetic energy lost by the car. [1]
- (b) Calculate the kinetic energy of the car before braking. [2]
- (c) Calculate the stopping distance. [2]
- (d)  The car now travels at twice the speed and brakes with the same force. The stopping distance becomes: the same, twice as long, four times as long, eight times as long. Circle the correct answer. [1]

Formula sheet and constants

Speed

$$v = \frac{s}{t}$$

Weight

$$W_{\text{weight}} = mg$$

Gravitational potential energy

$$\Delta E_p = mg \Delta h$$

Kinetic energy

$$E_k = \frac{1}{2} mv^2$$

Work

$$W = Fd \quad W = \Delta E$$

$$g = 10 \text{ N/kg}$$

$$1 \text{ J} = 1 \text{ N m}$$

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ h} = 3600 \text{ s}$$